LIBRARY MANAGEMENT SYSTEM

SQL Script Execution File | MySQL 8+

PART 1: SCHEMA DEFINITION & DATA INSERTION

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      -- LIBRARY MANAGEMENT SYSTEM: MAIN SQL PROGRAM -- Engine: MySQL 8+ CREATE
DATABASE IF NOT EXISTS LibraryManagementSystem; USE LibraryManagementSystem; -- 1.
CREATE TABLES CREATE TABLE IF NOT EXISTS Books ( BookID INT PRIMARY KEY, Title
VARCHAR(100) NOT NULL, Author VARCHAR(50) NOT NULL, Category VARCHAR(30) NOT NULL,
CopiesAvailable INT NOT NULL DEFAULT 1 ); CREATE TABLE IF NOT EXISTS Members (
MemberID INT PRIMARY KEY, MemberName VARCHAR(50) NOT NULL, Email VARCHAR(50) UNIQUE
NOT NULL, JoinDate DATE NOT NULL ); CREATE TABLE IF NOT EXISTS IssuedBooks ( IssueID
INT PRIMARY KEY, BookID INT NOT NULL, MemberID INT NOT NULL, IssueDate DATE NOT NULL,
DueDate DATE NOT NULL, ReturnDate DATE DEFAULT NULL, FOREIGN KEY (BookID) REFERENCES
Books(BookID), FOREIGN KEY (MemberID) REFERENCES Members(MemberID) ); -- 2. INSERT
SAMPLE DATA INSERT INTO Books (BookID, Title, Author, Category, CopiesAvailable)
VALUES (101, 'Database System Concepts', 'Silberschatz', 'Computer Science', 5), (102,
'Let Us C', 'Yashavant Kanetkar', 'Programming', 3), (103, 'Operating System
Concepts', 'Galvin', 'Computer Science', 2), (104, 'Malgudi Days', 'R.K. Narayan',
'Fiction', 4); INSERT INTO Members (MemberID, MemberName, Email, JoinDate) VALUES (1,
'Arun', 'arun@example.com', '2025-01-10'), (2, 'Priya', 'priya@example.com', '2025-02-
15'), (3, 'Kumar', 'kumar@example.com', '2025-03-01'), (4, 'Divya',
'divya@example.com', '2025-03-10');
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PART 2: TRANSACTIONS & QUERIES

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      INSERT INTO IssuedBooks (IssueID, BookID, MemberID, IssueDate, DueDate,
ReturnDate) VALUES (1, 101, 1, '2026-10-01', '2026-10-15', '2026-10-14'), (2, 102, 2,
'2026-10-05', '2026-10-19', NULL), (3, 103, 3, '2026-09-20', '2026-10-04', NULL), (4,
104, 4, '2026-10-10', '2026-10-24', '2026-10-22'); -- 3. DISPLAY ALL RECORDS SELECT *
FROM Books; SELECT * FROM Members; SELECT * FROM IssuedBooks; -- 4. DISPLAY CURRENTLY
ISSUED BOOKS WITH DETAILS SELECT I.IssueID, M.MemberName, B.Title, I.IssueDate,
I.DueDate FROM IssuedBooks AS I JOIN Books AS B ON I.BookID = B.BookID JOIN Members AS
M ON I.MemberID = M.MemberID WHERE I.ReturnDate IS NULL; -- 5. OVERDUE BOOKS AND FINE
CALCULATION (Rs. 5 per day) SELECT I.IssueID, M.MemberName, B.Title, I.DueDate,
CURRENT_DATE() AS CurrentDate, DATEDIFF(CURRENT_DATE(), I.DueDate) AS DaysOverdue,
(DATEDIFF(CURRENT_DATE(), I.DueDate) * 5) AS FineAmount FROM IssuedBooks AS I JOIN
Books AS B ON I.BookID = B.BookID JOIN Members AS M ON I.MemberID = M.MemberID WHERE
I.ReturnDate IS NULL AND I.DueDate < CURRENT_DATE();
```